UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/535,786	12/11/2023	Wonjong CHO	59163-0145001	2085

26191 7590 06/02/2026
FISH & RICHARDSON P.C. (TC)
PO BOX 1022
MINNEAPOLIS, MN 55440-1022

EXAMINER	
POWERS, LAURA C	

ART UNIT	PAPER NUMBER
1785	

NOTIFICATION DATE	DELIVERY MODE
06/02/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

PATDOCTC@fr.com

Office Action Summary	Application No. 18/535,786	Applicant(s) CHO et al.	
	Examiner LAURA POWERS	Art Unit 1785	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 13 April 2026.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-21 is/are pending in the application.

5a) Of the above claim(s) 11-21 is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-10 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 11 December 2023 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 11/20/2025.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Election/Restrictions

2. Applicant's election without traverse of Group I, claims 1-10, in the reply filed on 04/13/2026 is acknowledged.

3. Claims 11-21 are withdrawn from further consideration pursuant to 37 CFR 1.142(b) as being drawn to a nonelected invention, there being no allowable generic or linking claim. Election was made **without** traverse in the reply filed on 04/13/2026.

Priority

4. Receipt is acknowledged of certified copies of papers required by 37 CFR 1.55.

Information Disclosure Statement

5. The information disclosure statement (IDS) submitted on 11/20/2025 is considered by the examiner.

Claim Rejections - 35 USC § 103

6. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

7. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

8. The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

9. Claims 1, 2, 3, 4, 5, 6, 7, 8 and 10 are rejected under 35 U.S.C. 103 as being unpatentable over Shimizu et al. (WO 2019/225183A1, machine translation via EPO).

10. Regarding claims 1, 6 and 8, Shimizu et al. teaches a circular polarizer (light control film) used with display devices, comprised of a polarizer, a phase difference layer (obliquely-stretched base) and an adhesion layer (horizontally oriented dye layer) ([0005, 0009-0014, 0019-0026]). The phase difference layer (obliquely-stretched base) is comprised of a stretched film, such as an acrylic film or cyclic olefin film, wherein one embodiment stretches the film obliquely ([0021-0028]). The adhesion layer (horizontally oriented dye layer) is comprised of dye compounds and an acrylic adhesive (horizontal alignment film integrated with dye) comprising (meth)acrylic polymers) as the base polymer and containing alkyl (meth) acrylate as the main monomer unit ([0012-0018, 0063-0066]).

11. Shimizu et al. does not expressly teach that the circular polarize (light control film) comprises an anti-reflection layer. Park et al., in the same field of endeavor, teaches a variable transmittance device used in display devices, comprised of a first polarizer, a lower substrate, a lower alignment layer, an upper substrate, an upper alignment layer and a second polarizer, wherein the variable transmittance device reduces color non-uniformity in display devices ([0004-0005, 0013-0023, 0034-0043, 0045-0053, 0058-0063, 0068-0074]). Park et al. teaches that the variable transmittance device can further comprise an anti-reflection layer attached to the outside of the variable transmittance device via an adhesive, wherein anti-reflection layers are known

in the art, thus an appropriate anti-reflection layer and thickness can be appropriately selected based on the purpose of the application.

12. It would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify the circular polarize (light control film) taught by Shimizu et al. to attach the anti-reflection layer taught by Park et al. to the adhesion layer (horizontally oriented dye layer) to provide anti-reflective features to the circular polarize (light control film) based on the desired use and properties for the circular polarize (light control film), as Park et al. teaches such film are known in the art and one of ordinary skill in the art would be capable of selecting the appropriate anti-reflection film based on the purpose of the circular polarize (light control film).

13. **Regarding claim 2 and 3,** Shimizu et al. and Park et al. teach all the limitations of claim 1 above.

14. Shimizu et al. teaches that the in-plane phase difference (Re) refers to the in-plane phase difference measure with light of a certain wavelength at 23˚C, wherein the in-plane phase difference (Re) of the phase difference layer (obliquely-stretched base) satisfies the equation 115nm ≤ Re(550) ≤ 135nm ([0019-0026]).

15. The instant claims do not recite or require a wavelength at which the claimed plane direction phrase retardation value (Rin) is measured at to obtained the claimed ranges. Shimizu et al. teaches a phase difference layer (obliquely-stretched base) having the same structure and materials as the claimed obliquely stretched base (see pg-pub of instant application [0064-0065]), therefore, the phase difference layer (obliquely-stretched base) taught by Shimizu et al. would inherently have a similar plane direction phrase retardation value (Rin) to that of claims 2 and 3.

16. **Regarding claims 4 and 5,** Shimizu et al. and Park et al. teach all the limitations of claim 1 above. Shimizu et al. further teaches that the angle between the absorption axis of the polarizer and the slow phase axis of the phase difference layer (obliquely-stretched base) is between 39˚ and 51˚ and that the phase difference layer (obliquely-stretched base) has a slow axis in the oblique direction, for example a direction of 45˚ ([0010, 0034, 0038, 0042, 0082]).

17. **Regarding claim 7,** Shimizu et al. and Park et al. teach all the limitations of claim 1 above. Shimizu et al. further teaches dye compounds used in the adhesion layer (horizontally oriented dye layer), wherein the dye compounds can be organic or inorganic dye compounds, and can be used individually of in combinations of two or more types ([0012-0018]).

18. While Shimizu et al. does not expressly teach that the dyes comprise three types of dyes selected from among dye red, dye green, dye blue, dye cyan, dye magenta or dye yellow, such a modification would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention as an obvious matter of design choice based upon the desired visual effects of the resultant circular polarizer (light control film). Shimizu et al. teaches a number of examples of dyes and, as previously stated, teaches that dyes can be used individually or in combinations of two or more types ([0012-0018]). It is well within the level of one having ordinary skill in the art to choose three colors of dye from those listed in claim 7 as an obvious matter of design choice based upon the colors preferred by the user.

19. **Regarding claim 10,** Shimizu et al. and Park et al. teach all the limitations of claim 1 above. Shimizu et al. further teaches that the amount of dye compound in the

adhesion layer (horizontally oriented dye layer) is preferably 0.01 to 10 parts by weight per 100 parts by weight of the adhesion layer (horizontally oriented dye layer) to suppress the decrease in visible light transmittance of the circular polarizer (light control film) while bringing the reflective hue of the circular polarizer (light control film) closer to neutral ([0017]).

20. While Shimizu et al. does not expressly teach that the single transmittance of the adhesion layer (horizontally oriented dye layer) ranges from 60-90%, such a modification would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to adjust the transmittance of the adhesion layer (horizontally oriented dye layer) based upon the desired amount of transmittance of the adhesion layer (horizontally oriented dye layer) in the circular polarizer (light control film)

21. Claim 9 is rejected under 35 U.S.C. 103 as being unpatentable over Shimizu et al. (WO 2019/225183A1, machine translation via EPO) in view of Part et al. (KR 2021-0032699A, cited on IDS, machine translation via EPO provided) and further in view of Nagayama et al. (US 20022/0300103).

22. **Regarding claim 9,** Shimizu et al. and Park et al. teach all the limitations of claim 1 above, however, the references do not expressly teach that the adhesion layer (horizontally oriented dye layer) comprises a photoinitiator.

23. Nagayama et al. teaches a resin composition and adhesive member used with a light control layer and a display device, wherein the adhesive layer is formed by photocuring a resin composition and exhibits excellent adhesion at high temperatures

resulting in excellent reliability when used in flexible display devices (Abstract, [0006-0020, 0027]). The adhesive is comprised of a (meth)acrylic monomer and as photoinitiator activated by ultraviolet light ([0086-0092, 0105-0108]).

24. As both Shimizu et al. and Nagayama et al. teach adhesive layers comprising (meth)acrylic polymers used with light control layers/polarizers, it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention, to modify the adhesion layer (horizontally oriented dye layer) taught by Shimizu et al. to include a photoinitiator as taught by the adhesive layer of Nagayama et al., as Nagayama et al. teaches that such adhesive layers exhibit excellent adhesion at high temperatures and excellent reliability when used in display devices.

Conclusion

25. Any inquiry concerning this communication or earlier communications from the examiner should be directed to LAURA POWERS whose telephone number is (571)270-5624. The examiner can normally be reached Monday-Thursday, 10:00AM-3:00PM.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Mark Ruthkosky can be reached at 571-272-1291. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

LAURA POWERS
Examiner
Art Unit 1785

/LAURA C POWERS/
Primary Examiner, Art Unit 1785